Analysis Report

Review of FlowSOM clustering and UMAP to identify populations

2026-11-13

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1 Overall summary

This run **15 FlowSOM metaclusters** were created from **5 Patient Samples.**

Below, the number of cells analyzed per sample and the top **3** clusters can be observed:

Table 1: Total cells and top represented clusters per sample

Sample	Cells per sample, cleaned	Top represented clusters
11 PT169 immune panel	21658	Metacluster 8 (77.4%); Metacluster 6 (8.0%); Metacluster 3 (5.8%)

Sample	Cells per sample, cleaned	Top represented clusters
13 PT170	22620	Metacluster 4 (51.2%);
immune		Metacluster 3 (17.9%);
panel		Metacluster 6 (15.3%)
15 PT171	10031	Metacluster 12 (36.6%);
immune		Metacluster 4 (36.2%);
panel		Metacluster 6 (11.7%)
17 PT172	12938	Metacluster 8 (55.9%);
immune		Metacluster 12 (14.3%);
panel		Metacluster 6 (11.2%)
9 PT160	19040	Metacluster 3 (24.8%);
immune		Metacluster 6 (16.3%);
panel		Metacluster 4 (13.3%)

1.1 Representation of events per metacluster

This may help determine if the samplers were "overclustered" resulting in clusters with very low number of cells that may not be relevant.

2 Identify Metaclusters

The images below are to help identify and eventually annotate the clusters.

3 Metacluster Representation in Patients

Let's visualize the clusters on patients

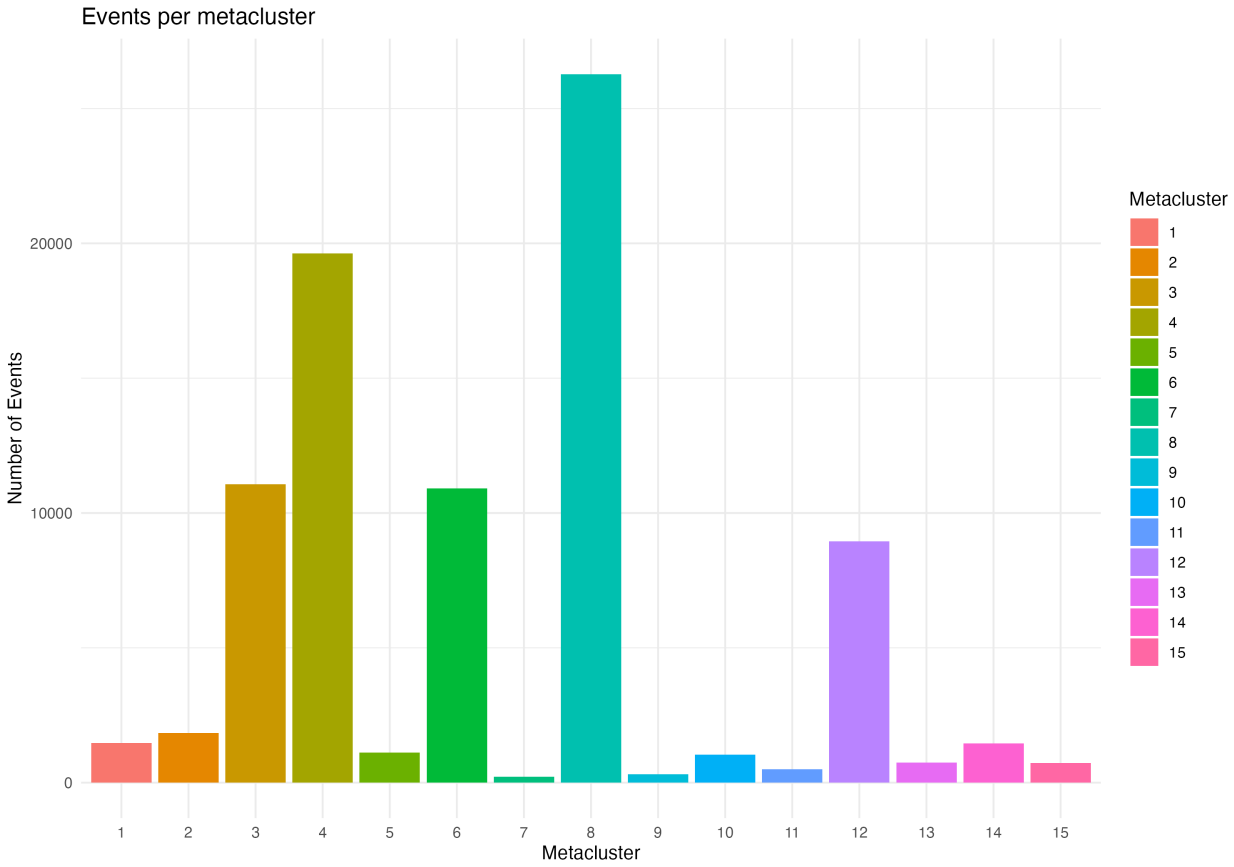


Figure 1: Events per Metacluster

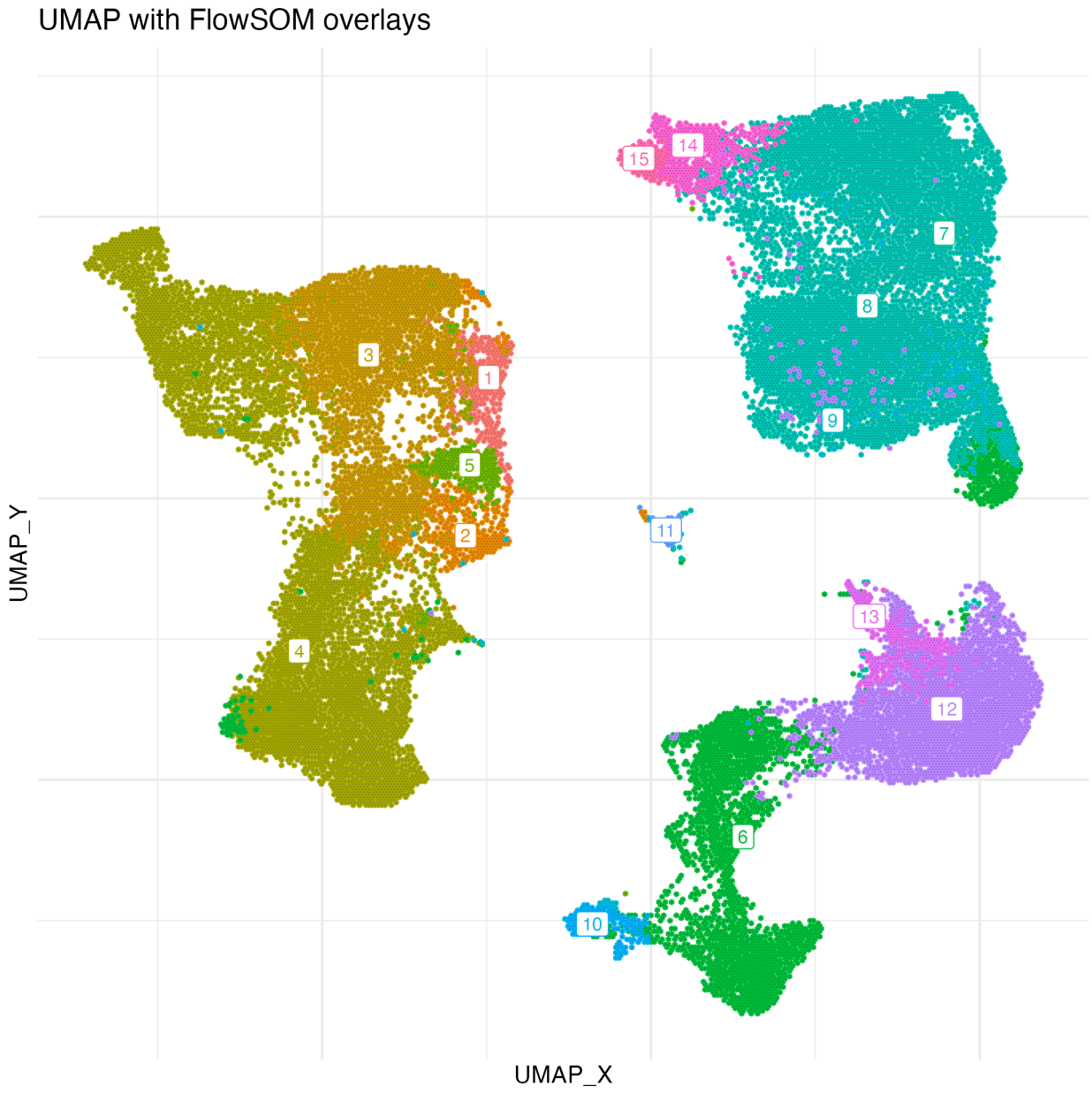


Figure 2: UMAP with FlowSOM Metacluster overlay

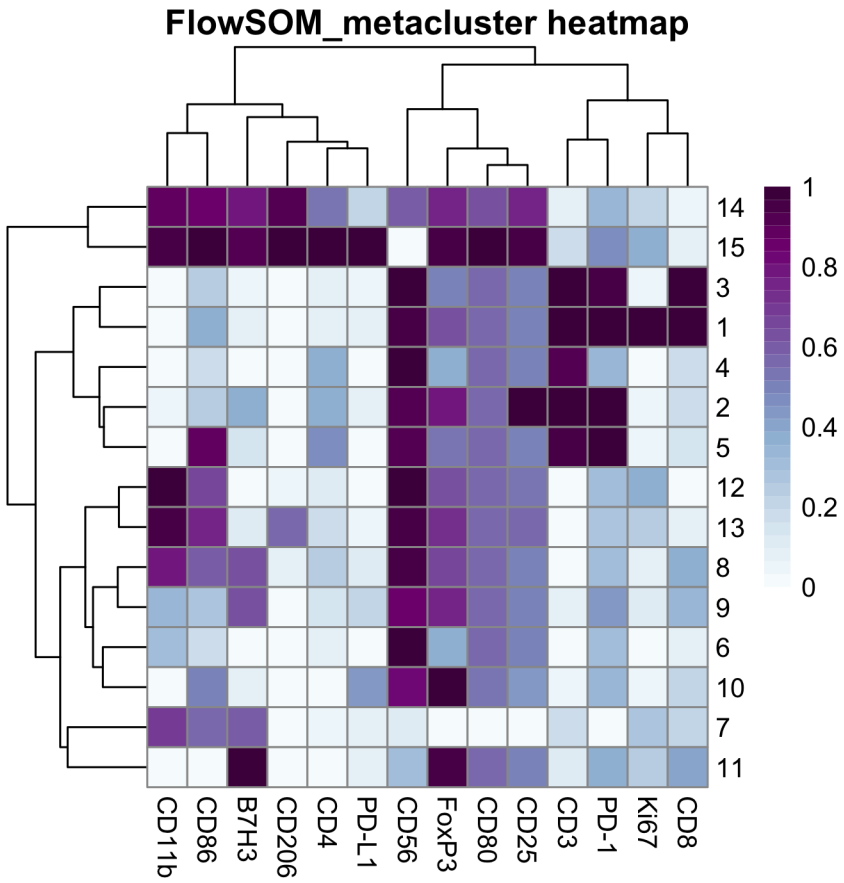


Figure 3: Expression Heatmap of Antigens against Metaclusters

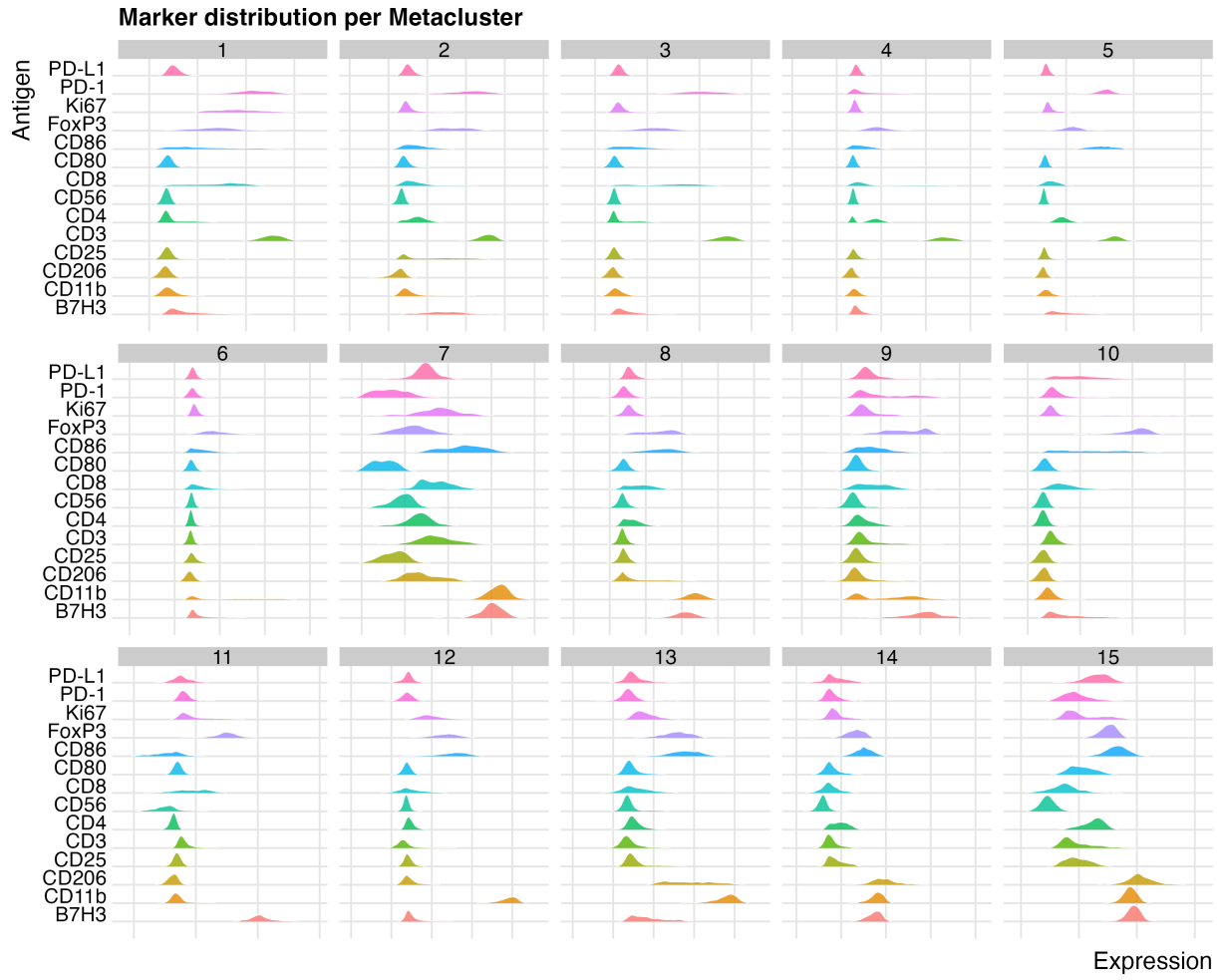


Figure 4: Antigen Expression per Metacluster

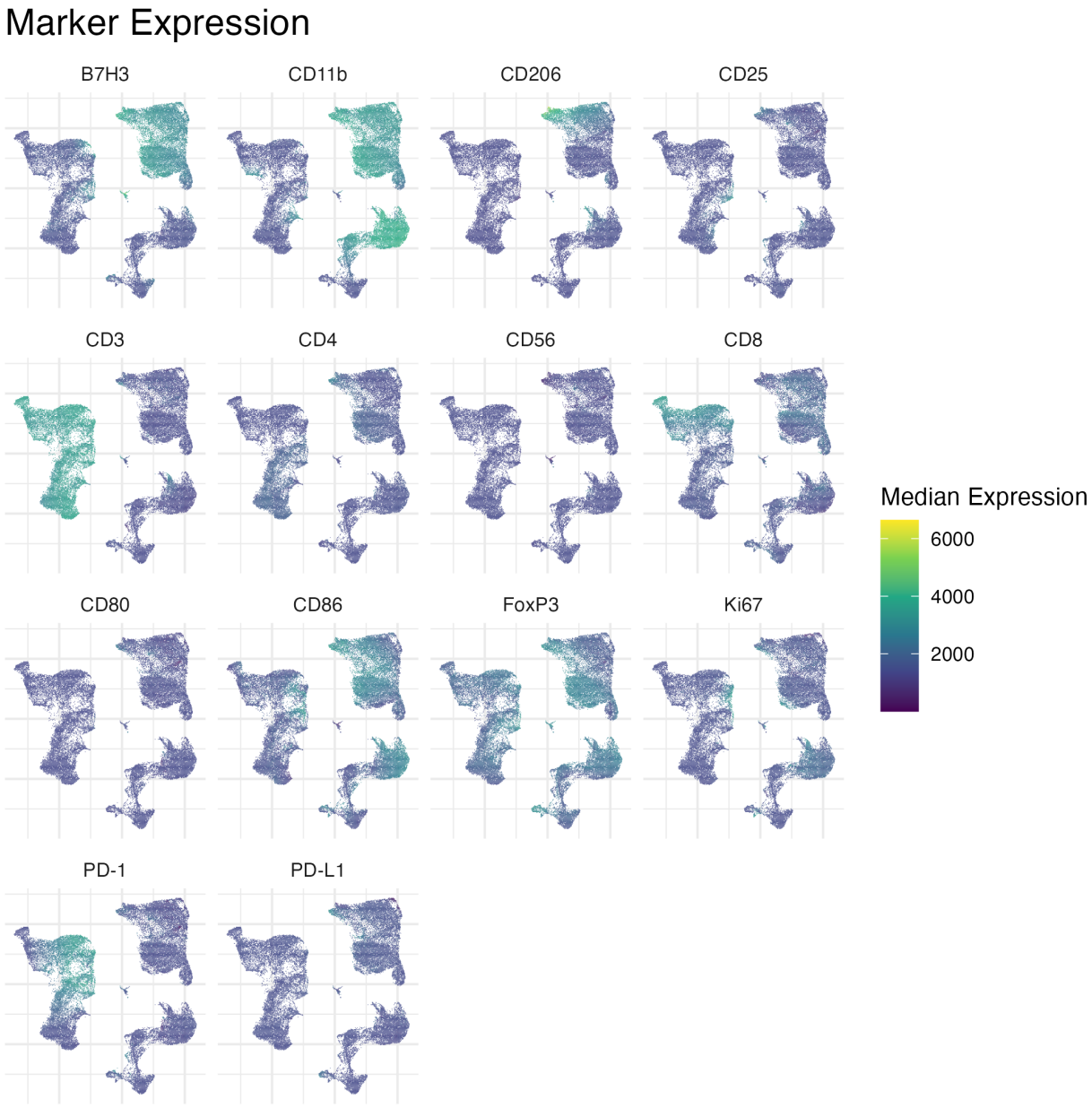


Figure 5: UMAP with Antigen Expression Overlaid

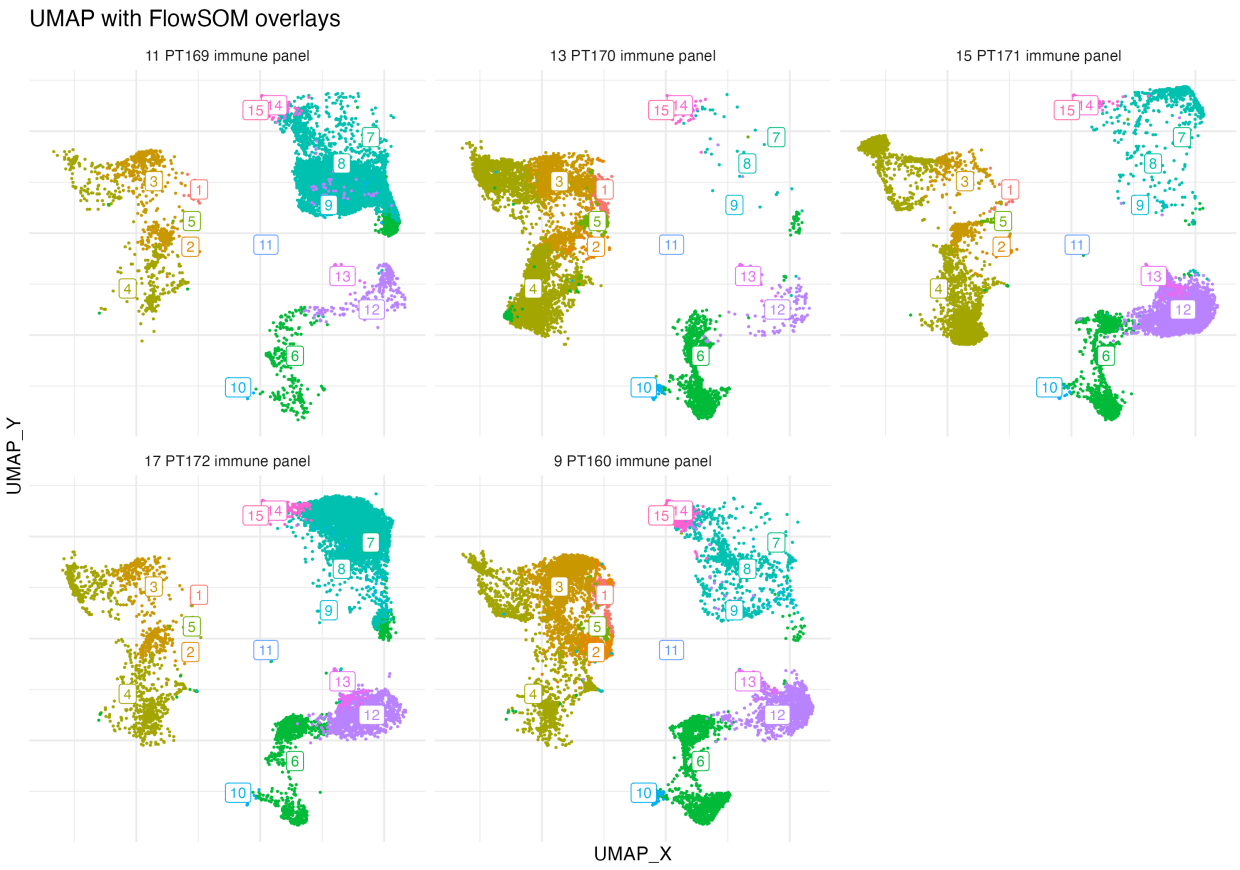


Figure 6: UMAP per patient

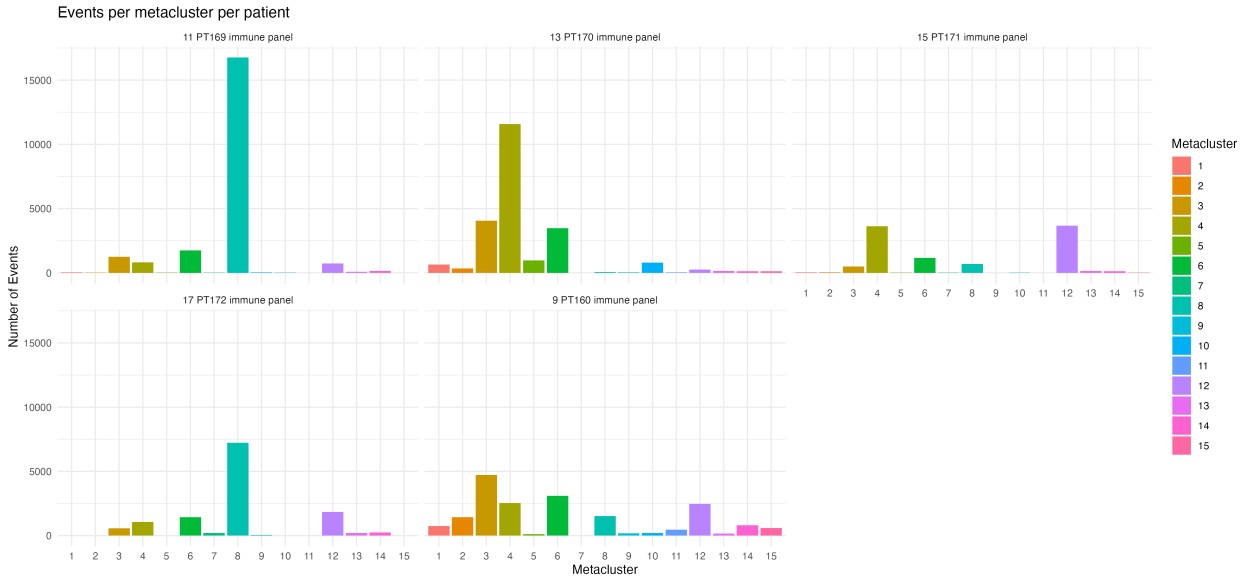


Figure 7: Events per Metacluster per patient